SYLVIE YU

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SUMMARY OF QUALIFICATIONS

University of Washington biostatistics master's student with Pfizer data science experience in patient retention modeling, pharmaceutical sales forecasting, and automated reporting. Applied Python, R, and SAS across industry and biomedical projects, including survival analysis, machine learning, lipidomics, and a mentored clinical trial analysis. Interested in biostatistics and data science roles in pharmaceuticals and healthcare.

EDUCATION

University of Washington *Sep 2026 – Mar 2028 (Expected)*
M.S. in Biostatistics, Data Science Track Seattle, WA
Relevant courses:

Shanghai Jiao Tong University *Sep 2022 – Jun 2026*
Bachelor of Science (B.S.) in Medical Laboratory Science Shanghai, China

PROFESSIONAL EXPERIENCE

Pfizer – GCA/DS, Commercial Data Analytics Team *Feb 2026 – Jun 2026*
Data Science Intern Shanghai, China

- Applied **survival analysis** and **predictive modeling** to real-world patient support program data, building Cox proportional hazards and accelerated failure time (AFT) models in **Python** to analyze patient retention; integrated findings from Kaplan–Meier, Cox-family, AFT, and **machine learning (ML)** analyses, including LightGBM, to assess consistent patient drop-off risk signals, and evaluated alternative tree-based models for predictive performance.
- Developed a reusable Python command-line workflow (CLI) to prepare data and consolidate analytical outputs, and designed detailed prompts incorporating analytical procedures and results for LLM-assisted generation of HTML reports for business audiences; deployed the workflow in **Dataiku**, collaborated with module owners to validate outputs, and authored comprehensive SOPs, reusable AI skill instructions, and project documentation for standardized reruns and handover.
- Addressed a recent sales spike associated with a sales-driven inventory push in a pharmaceutical **time series forecasting project**; estimated the anomalous month's baseline sales using earlier historical data, then blended the estimate with observed sales using empirically selected weights to replace the modeling input while preserving monthly continuity.
- Reduced average monthly forecast error **from 17% to 3%** at the franchise level in project evaluations through iterative anomaly adjustment, feature engineering, model comparisons, and post-processing. Tested AutoARIMA, hierarchical forecasting, and ensembles while assessing predictive performance and overfitting.
- Investigated missing data, delayed refreshes, and chatbot issues in an internal data assistant; checked database records and communicated findings to help distinguish source-data, pipeline, and application problems.

Xinhua Hospital *May 2025 – Nov 2025*
Clinical Laboratory Science Intern Shanghai, China

- Completed six months of clinical laboratory rotations across hematology, biochemistry, immunology, microbiology, and molecular diagnostics, gaining firsthand exposure to laboratory quality control, biomarker measurement, and sources of pre-analytical and analytical variation.

PROJECT EXPERIENCE

Lipoprotein Subfraction and Lipidomic Profiling in Acute Myocardial Infarction *Nov 2025 – Apr 2026*
Shanghai Tongren Hospital (Shanghai Jiao Tong University School of Medicine), Supervised by Dr. Qian Wu

- Cleaned clinical data for **50 participants** (25 myocardial infarction patients and 25 healthy controls) in **R**. Generated descriptive statistics and baseline comparisons using t-tests, Mann–Whitney U tests, chi-square tests, and Fisher's exact tests as appropriate.
- Compared lipoprotein subfractions and approximately 25 free fatty acids selected under predefined criteria using Welch's t-tests and Benjamini–Hochberg false discovery rate correction. Reviewed mass-spectrometry chromatograms and peak integration in instrument software; curated and analyzed quantitative data in R. Compared long-chain and very-long-chain fatty-acid class totals, with Wilcoxon rank-sum robustness checks.
- Screened untargeted lipidomics using quality-control variability, supplied variable importance in projection scores, fold changes, and statistical significance. Combined positive- and negative-ion datasets and reviewed metabolite annotations to prioritize lipid features.
- Integrated lipoprotein, fatty-acid, and untargeted lipidomic results using Spearman and Pearson correlations with multiple-testing correction. Created boxplots and correlation heatmaps in R; wrote the literature review and thesis, interpreting medium-sized low-density lipoprotein (LDL) patterns as exploratory given the small observational sample.

SAS Clinical Programming and Statistical Analysis — Phase III Acne Trial *Feb 2025 – May 2025*
Independent Clinical Statistics Training Project | Mentored by an IQVIA statistician Shanghai, China

- Analyzed historical data from an approximately **1,000-participant** randomized, double-blind, placebo-controlled trial in **SAS** during a guided training project; summarized demographic and baseline characteristics.
- Programmed co-primary efficacy analyses: **analysis of covariance (ANCOVA)** with last observation carried forward for Week 12 inflammatory lesion-count changes, and an age-stratified **Cochran–Mantel–Haenszel tes**t for Investigator's Global Assessment treatment success.
- Summarized adverse events by treatment group and heart-rate changes; prepared statistical tables and a final presentation explaining study design, endpoints, methods, and efficacy and safety findings.

Colorectal Cancer Transcriptome Analysis & Immune-Related Prognostic Modeling *Jun 2023 – Sep 2024*
Shanghai Jiao Tong University School of Medicine, Supervised by Dr. Dakang Xu

- Independently learned **R** to reproduce the analytical workflow of a published colorectal cancer bioinformatics study using **500+ transcriptomic profiles** from TCGA/GEO.
- Performed differential expression analysis, KEGG pathway enrichment, and ESTIMATE scoring, alongside Kaplan–Meier analysis, log-rank tests, and Cox proportional hazards modeling, to reproduce the study's analyses of tumor biology and survival associations.

Bile Acid Metabolism, Cognition, and Schizophrenia Progression *Oct 2022 – May 2024*
Shanghai Jiao Tong University, Supervised by Ying Qing and Lihua Xu

- Co-developed the hypothesis and led the design of a three-group case-control study involving individuals at clinical high risk for psychosis, patients with first-episode schizophrenia, and healthy controls; used ANOVA to compare seven cognitive domains and 36 bile acids across groups and logistic regression to examine bile-acid–cognition associations.

TECHNICAL SKILLS

Programming & Tools: Python (intermediate), R (intermediate), SAS (intermediate), Microsoft Excel, DBeaver, Dataiku
Statistical & Data Analysis: Data cleaning and visualization, exploratory data analysis, survival analysis, time series forecasting, machine learning, model evaluation, hypothesis testing, ANCOVA, Cochran–Mantel–Haenszel analysis, correlation analysis, multiple testing correction, LLM-assisted reporting and workflow automation